# Logistic Regression

## Part 3a: binary classification. Part 3b: multiclass classification with softmax regression

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# This lecture

In classification the target is a **category**, not a number. Part 3a treats two classes, part 3b any number of classes; the second contains the first as the special case $K = 2$.

**Part 3a: binary classification**

- why linear regression fails for classification;
- the **logistic regression** model, with outputs that are **probabilities**;
- **decision boundaries** (linear and nonlinear) and the **threshold**;
- the **logistic loss**, and why not the squared error;
- **gradient descent**, and logistic regression in **scikit-learn**.

**Part 3b: multiclass classification**

- multiclass problems and their notation;
- the **softmax** function and **softmax regression**;
- decision regions for $K$ classes;
- the **cross-entropy loss**, its gradient and gradient descent;
- softmax regression in **scikit-learn**, the **confusion matrix**, and **one-vs-rest**.

# Part 3a: binary classification

# Classification

# Classification problems

| question | answer |
|---|---|
| Is this email spam? | no / yes |
| Is this transaction fraudulent? | no / yes |
| Is this tumour malignant? | no / yes |

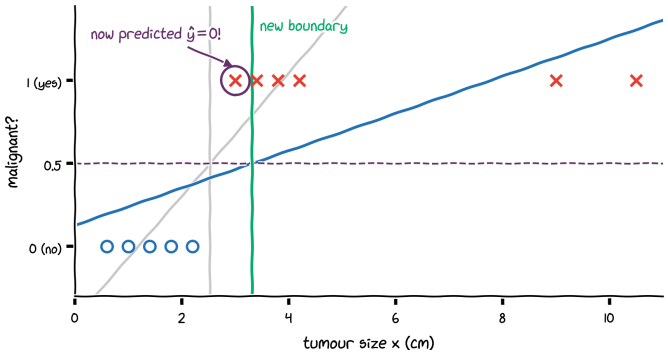
- The target $y$ can take only **two** values: **binary classification**.
- Each value is a **class** (category).

We encode the two classes as numbers:

| answer | $y$ | name |
|---|---|---|
| no, false | 0 | **negative** class |
| yes, true | 1 | **positive** class |

"Negative" and "positive" do not mean bad and good. The positive class is the **presence** of the property we are looking for (spam, fraud, a malignant tumour); the negative class is its absence.

# Why not linear regression?




...but the least-squares line tilts, the boundary moves from 2.53 to 3.31 cm, and a malignant tumour is now predicted benign.

# What we need instead

The linear-regression approach has two problems:

- **Easy examples move the boundary.** Points far from the boundary, which are easy to classify, pull the line and shift the boundary.
- **The output is not a probability.** $f_{w,b}(x) = wx + b$ can be below 0 or above 1, so it cannot be read as "the chance that the tumour is malignant".

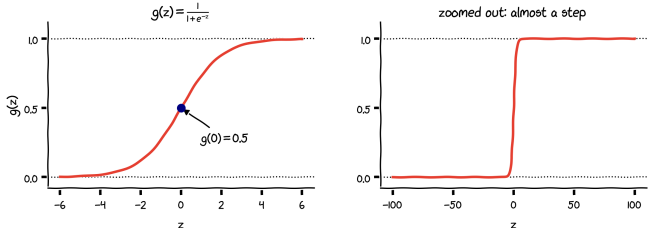
We want a model whose output

- always lies between 0 and 1, and can be read as a probability;
- is close to 0 or 1 far from the boundary, so easy examples have little influence.

This is **logistic regression**: despite its name, it is a **classification** algorithm.

# The logistic regression model

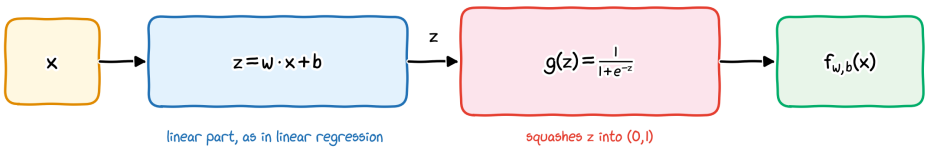
# The sigmoid (logistic) function




$$g(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+\exp(-z)}$$

- $0 < g(z) < 1$ for every $z$, and $g(0) = 0.5$.
- $g(z) \to 1$ as $z \to \infty$ and $g(z) \to 0$ as $z \to -\infty$; e.g. $g(10) \approx 0.99995$.
- Symmetry: $g(-z) = 1 - g(z)$.

# The logistic regression model

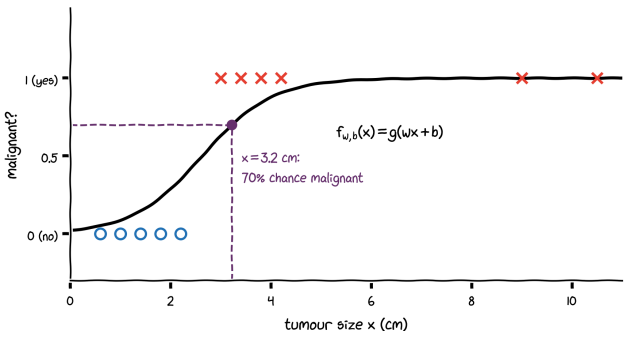



**Logistic regression**

$$f_{\mathbf{w},b}(\mathbf{x}) = g(\mathbf{w} \cdot \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w} \cdot \mathbf{x} + b)}}$$

- The **linear part** $z = \mathbf{w} \cdot \mathbf{x} + b$ is exactly the model of Lecture 2.
- The **sigmoid** squashes $z$ into the interval $(0, 1)$.

# Logistic regression on the tumour data




The same eleven patients, including the two large tumours.

- The sigmoid saturates at 1 for large tumours, so they hardly affect the fit.
- Fitted with scikit-learn: $w = 1.44$, $b = -3.77$. The boundary $f = 0.5$ lies at $x = 2.63$ cm, between the two classes.
- For a tumour of 3.2 cm the model gives $f = 0.7$: a **70% chance** of being malignant.

# The output is a probability

$$f_{\mathbf{w},b}(\mathbf{x}) = P(y = 1 \mid \mathbf{x};\ \mathbf{w}, b)$$

"The probability that $y = 1$, given the input $\mathbf{x}$, for parameters $\mathbf{w}$ and $b$."

- The two probabilities add up to one: $P(y = 0 \mid \mathbf{x}) = 1 - P(y = 1 \mid \mathbf{x}) = 1 - f_{\mathbf{w},b}(\mathbf{x})$.
- Example: $f_{\mathbf{w},b}(\mathbf{x}) = 0.7$ means a 70% chance that the tumour is malignant and a 30% chance that it is benign.

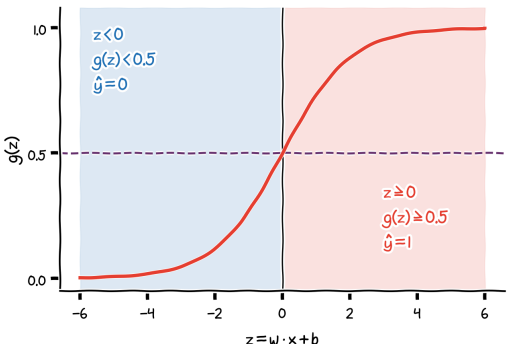
**Optional: log-odds.** Solving $f = g(z)$ for $z$ gives

$$z = \mathbf{w} \cdot \mathbf{x} + b = \log \frac{f}{1 - f} \qquad \text{(the log-odds of class 1).}$$

Increasing $x_j$ by one unit adds $w_j$ to the log-odds, i.e. multiplies the odds $f/(1 - f)$ by $e^{w_j}$.

# Decision boundary

# From probabilities to predictions




To predict a class, compare the probability with a **threshold**, usually 0.5:

$$\hat{y} = \begin{cases} 1 & \text{if } f_{\mathbf{w},b}(\mathbf{x}) \geq 0.5 \\ 0 & \text{if } f_{\mathbf{w},b}(\mathbf{x}) < 0.5 \end{cases}$$

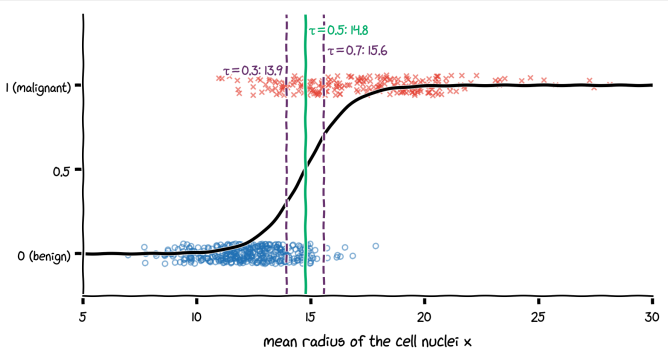
Since $g(z) \geq 0.5$ exactly when $z \geq 0$:

$$\hat{y} = 1 \iff \mathbf{w} \cdot \mathbf{x} + b \geq 0.$$

The **decision boundary** is the set of points where

$$\mathbf{w} \cdot \mathbf{x} + b = 0.$$

# Decision boundary in 1D: real data




Breast-cancer data: 569 tumours, feature $x$ = mean radius of the cell nuclei. With one feature the boundary is a single point, $wx + b = 0 \iff x = -b/w$. Here $w = 1.034$, $b = -15.25$, so $x = 14.75$.

Data: Street, Wolberg & Mangasarian (1993), Wisconsin breast cancer data; in Python `sklearn.datasets.load_breast_cancer`.

**Question**: what happens if we change the threshold from 0.5 to 0.7?

# Changing the threshold

For a threshold $\tau$ with $0 < \tau < 1$, the boundary is where $f = \tau$:

$$\frac{1}{1+e^{-z}} = \tau \iff e^{-z} = \frac{1}{\tau} - 1 \iff z = \log\frac{\tau}{1-\tau}.$$

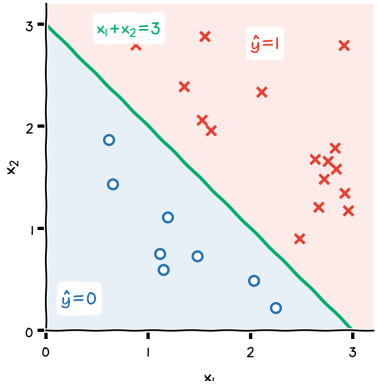
In 1D, with $z = wx + b$:

$$x = \frac{1}{w}\left(\log\frac{\tau}{1-\tau} - b\right).$$

For $\tau = 0.5$ this gives $\log 1 = 0$ and $x = -b/w$ as before. For the breast-cancer data, $\tau = 0.3, 0.5, 0.7$ give $x = 13.9,\ 14.75,\ 15.6$.

- A **higher** $\tau$ predicts "malignant" only when the model is more certain: fewer false alarms, but more missed tumours.
- A **lower** $\tau$ catches more malignant tumours at the cost of more false alarms; in medical screening this is often preferred.

# Decision boundary in 2D




With two features

$$f_{\mathbf{w},b}(\mathbf{x}) = g(w_1 x_1 + w_2 x_2 + b),$$

and the decision boundary is the straight line

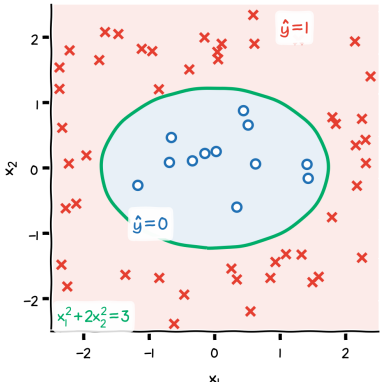
$$w_1 x_1 + w_2 x_2 + b = 0.$$

Example: after training, $w_1 = w_2 = 1$ and $b = -3$:

$$x_1 + x_2 - 3 = 0 \iff x_1 + x_2 = 3,$$

with $\hat{y} = 1$ where $x_1 + x_2 \geq 3$.
With $n$ features the boundary $\mathbf{w} \cdot \mathbf{x} + b = 0$ is a (hyper)plane.

# Nonlinear decision boundaries




Use polynomial features, as in polynomial regression:

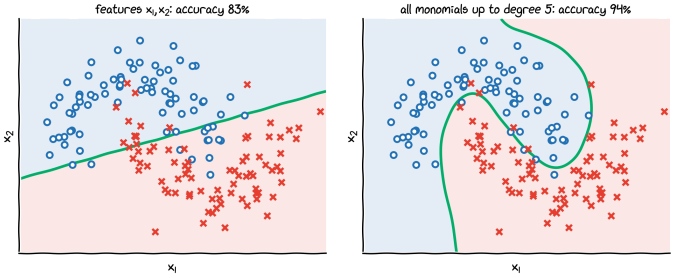
$$f_{\mathbf{w},b}(\mathbf{x}) = g(w_1 x_1^2 + w_2 x_2^2 + b).$$

Example: after training, $w_1 = 1$, $w_2 = 2$ and $b = -3$. The decision boundary is

$$x_1^2 + 2x_2^2 - 3 = 0 \iff x_1^2 + 2x_2^2 = 3,$$

an **ellipse**. Outside it $z \geq 0$, so $\hat{y} = 1$; inside it $\hat{y} = 0$.
Changing the sign of all parameters swaps the two regions.

# Even more flexible boundaries




- Left: only $x_1$ and $x_2$, so the boundary is a straight line.
- Right: all 20 monomials up to degree 5 ($x_1, x_2, x_1^2, x_1x_2, \ldots, x_2^5$). The boundary can take almost any shape.
- More flexible is not always better: a very wiggly boundary may follow the noise in the training data (**overfitting**, Lecture 4).

# The cost function

# The training set

| tumour size (cm) | ⋯ | age | malignant? |
|---|---|---|---|
| $x_1$ | ⋯ | $x_n$ | $y$ |
| 2.5 | | 52 | 1 |
| 0.8 | | 73 | 0 |
| 1.6 | | 55 | 0 |
| 3.1 | | 49 | 1 |
| ⋮ | | ⋮ | ⋮ |

- Training examples $i = 1, \dots, m$; features $j = 1, \dots, n$.
- The target $y^{(i)}$ is 0 or 1.
- Model:

$$f_{\mathbf{w},b}(\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w} \cdot \mathbf{x} + b)}}$$

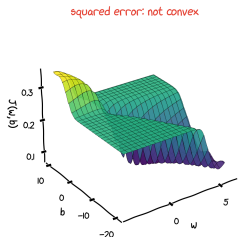
How do we choose $\mathbf{w}$ and $b$? We need a **cost function** that measures how well $f_{\mathbf{w},b}$ fits the training labels.

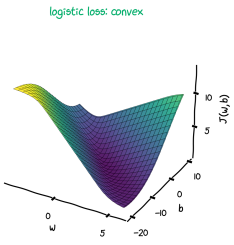
# Why not the squared error?

The cost of linear regression, written as an average of per-example **losses** $L$:

$$J(\mathbf{w}, b) = \frac{1}{m} \sum_{i=1}^{m} \underbrace{\frac{1}{2}\left(f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)}\right)^2}_{L\left(f_{\mathbf{w},b}(\mathbf{x}^{(i)}),\ y^{(i)}\right)}$$

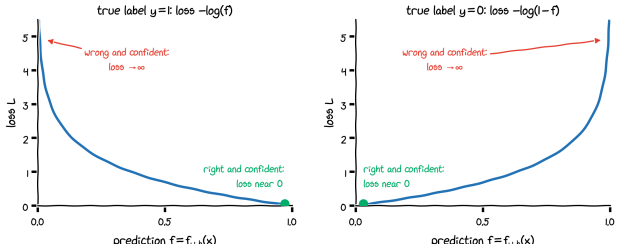
- Linear model $f = \mathbf{w} \cdot \mathbf{x} + b$: this $J$ is a convex bowl (Lecture 2).
- Sigmoid model (left plot): $J$ is **not convex**. It has flat plateaus where gradient descent stalls, and can have local minima.
- The logistic loss (right plot, next slides) gives a **convex** cost with a single minimum.



# The logistic loss

$$L\big(f_{\mathbf{w},b}(\mathbf{x}^{(i)}), y^{(i)}\big) = \begin{cases} -\log\big(f_{\mathbf{w},b}(\mathbf{x}^{(i)})\big) & \text{if } y^{(i)} = 1 \\ -\log\big(1 - f_{\mathbf{w},b}(\mathbf{x}^{(i)})\big) & \text{if } y^{(i)} = 0 \end{cases}$$




- The loss is 0 when the model predicts the true label with full confidence.
- It grows without bound when the model is confident **and wrong**: confident mistakes are punished severely.

# A single formula for the loss

Because $y^{(i)}$ is either 0 or 1, both cases fit into one expression:

$$L(f_{\mathbf{w},b}(\mathbf{x}^{(i)}), y^{(i)}) = -y^{(i)} \log\left(f_{\mathbf{w},b}(\mathbf{x}^{(i)})\right) - (1 - y^{(i)}) \log\left(1 - f_{\mathbf{w},b}(\mathbf{x}^{(i)})\right).$$

- If $y^{(i)} = 1$: $L = -1 \cdot \log(f) - 0 \cdot \log(1 - f) = -\log(f)$.
- If $y^{(i)} = 0$: $L = -0 \cdot \log(f) - 1 \cdot \log(1 - f) = -\log(1 - f)$.

**Cost function for logistic regression**

$$J(\mathbf{w}, b) = -\frac{1}{m} \sum_{i=1}^{m} \left[ y^{(i)} \log\left(f_{\mathbf{w},b}(\mathbf{x}^{(i)})\right) + (1 - y^{(i)}) \log\left(1 - f_{\mathbf{w},b}(\mathbf{x}^{(i)})\right) \right]$$

Note the minus sign in front of the sum. $J$ is **convex**, and training means solving $\min_{\mathbf{w},b} J(\mathbf{w}, b)$. Other names: **log loss**, **binary cross-entropy**.

# Where does the logistic loss come from? (optional)

The model states $P(y = 1 \mid \mathbf{x}) = f$ and $P(y = 0 \mid \mathbf{x}) = 1 - f$. Both cases in one formula:

$$P(y \mid \mathbf{x}) = f^{\,y}\,(1 - f)^{1-y}.$$

If the training examples are independent, the probability that the model assigns to all the observed labels (the **likelihood**) is

$$\prod_{i=1}^{m} P(y^{(i)} \mid \mathbf{x}^{(i)}).$$

Good parameters make this large. Taking $-\frac{1}{m}\log$ turns the product into a sum, and maximising becomes minimising:

$$-\frac{1}{m}\sum_{i=1}^{m} \log P(y^{(i)} \mid \mathbf{x}^{(i)}) = -\frac{1}{m}\sum_{i=1}^{m} \left[ y^{(i)} \log f^{(i)} + (1 - y^{(i)}) \log \left(1 - f^{(i)}\right) \right] = J(\mathbf{w}, b).$$

Minimising $J$ is **maximum likelihood estimation**: each loss is
$-\log$(probability of the true label).

# Gradient descent

# The gradient of the logistic cost

One example, with $f = g(z)$ and $z = \mathbf{w} \cdot \mathbf{x} + b$. The key fact about the sigmoid is

$$g'(z) = g(z)(1 - g(z)).$$

Chain rule, step by step:

$$\frac{\partial L}{\partial f} = -\frac{y}{f} + \frac{1-y}{1-f} = \frac{f-y}{f(1-f)}, \qquad \frac{\partial f}{\partial z} = f(1-f) \quad \Longrightarrow \quad \frac{\partial L}{\partial z} = f - y.$$

With $\partial z/\partial w_j = x_j$ and $\partial z/\partial b = 1$, and averaging over all examples:

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^{m} \left( f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)} \right) x_j^{(i)}, \qquad \frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^{m} \left( f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)} \right).$$

# Gradient descent for logistic regression

Repeat until convergence, updating all parameters **simultaneously**:

$$\begin{cases} w_j \leftarrow w_j - \alpha \dfrac{1}{m} \displaystyle\sum_{i=1}^{m} \left( f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)} \right) x_j^{(i)}, & j = 1, \dots, n \\ b \leftarrow b - \alpha \dfrac{1}{m} \displaystyle\sum_{i=1}^{m} \left( f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)} \right) \end{cases}$$

**Same form as linear regression...**

- linear regression: $f_{\mathbf{w},b}(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b$
- logistic regression:

$$f_{\mathbf{w},b}(\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w} \cdot \mathbf{x} + b)}}$$

...but a different $f$, so a different algorithm.

**The same practical tools apply**

- learning curves to monitor $J$;
- a vectorised implementation;
- feature scaling.

# Vectorised implementation

```
import numpy as np

def sigmoid(z):
    return 1.0 / (1.0 + np.exp(-z))

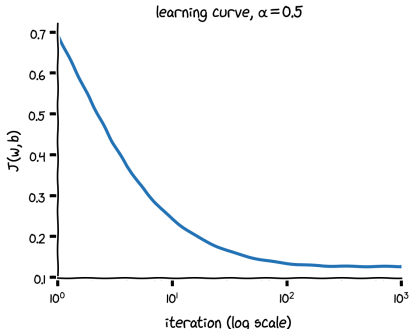
def cost(X, y, w, b):
    f = sigmoid(X @ w + b)
    return -np.mean(y * np.log(f)
                    + (1 - y) * np.log(1 - f))

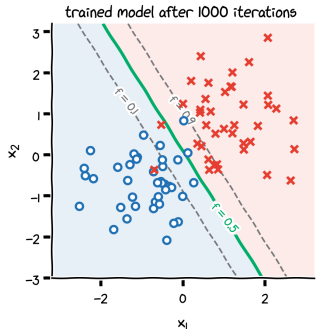
def gradient_descent(X, y, alpha=0.5, n_iters=1000):
    m, n = X.shape
    w, b = np.zeros(n), 0.0
    for _ in range(n_iters):
        f = sigmoid(X @ w + b)    # (m,)
        err = f - y               # (m,)
        w = w - alpha * X.T @ err / m
        b = b - alpha * err.mean()
    return w, b
```

Compared with linear regression, only one line changes: the predictions are `sigmoid(X @ w + b)` instead of `X @ w + b`.

On the two-class data of the next slide, 1000 iterations give $\mathbf{w} = (3.60,\ 2.26)$ and $b = -0.20$, the same as scikit-learn without regularisation.

# Gradient descent in action




- Left: $J$ decreases at every iteration and levels off, as it should.
- Right: the trained model. The boundary is the contour $f = 0.5$; between the dashed contours $f = 0.1$ and $f = 0.9$ the model is uncertain, and this is where the two classes overlap.

# Logistic regression with scikit-learn

# LogisticRegression in practice

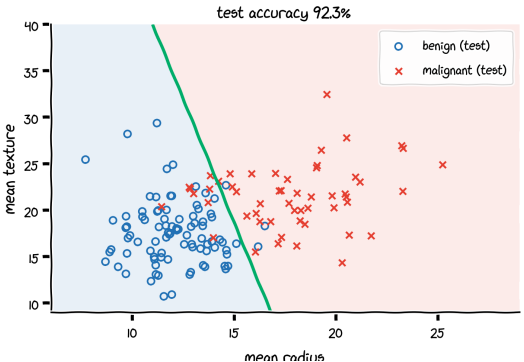
```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

data = load_breast_cancer()
X = data.data[:, :2]                    # mean radius, mean texture
y = 1 - data.target                     # 1 = malignant (scikit-learn codes it as 0)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=0, stratify=y)

model = make_pipeline(StandardScaler(), LogisticRegression())
model.fit(X_train, y_train)

model.predict(X_test[:3])               # array([1, 0, 0]): predicted classes
model.predict_proba(X_test[:3])         # rows [P(y=0), P(y=1)]: [0.04, 0.96], [0.89, 0.11], ...
model.score(X_test, y_test)             # 0.923: accuracy on the test set
```

# Results and remarks




- **Accuracy** = fraction of correct predictions. Two features: 92.3% on the test set; all 30 features: 98.6%.
- Always compare with a **baseline**: predicting "benign" for every tumour already gives 62.9%.
- `predict` uses the threshold 0.5; for another threshold, use `predict_proba(X)[:, 1] >= tau`.
- `LogisticRegression` applies **regularisation** by default ($C = 1$, Lecture 4). Use `C=np.inf` to switch it off (older versions: `penalty=None`).

# Summary of part 3a

- **Binary classification**: $y \in \{0, 1\}$; linear regression is a poor classifier.
- **Model**: $f_{\mathbf{w},b}(\mathbf{x}) = g(\mathbf{w} \cdot \mathbf{x} + b)$ with the sigmoid $g(z) = 1/(1 + e^{-z})$; the output is $P(y = 1 \mid \mathbf{x})$.
- **Prediction**: $\hat{y} = 1$ if $f \geq \tau$ (usually $\tau = 0.5$, i.e. $\mathbf{w} \cdot \mathbf{x} + b \geq 0$).
- **Decision boundary**: $\mathbf{w} \cdot \mathbf{x} + b = 0$ for $\tau = 0.5$; linear in the features, but nonlinear in $\mathbf{x}$ when we use polynomial features.
- **Cost**: $J = -\frac{1}{m} \sum_i \left[ y^{(i)} \log f^{(i)} + (1 - y^{(i)}) \log(1 - f^{(i)}) \right]$: convex, from maximum likelihood.
- **Gradient descent**: same update formulas as linear regression, with the sigmoid inside $f$.
- **scikit-learn**: `LogisticRegression`, `predict`, `predict_proba`, `score`; regularised by default.

**Next, part 3b**: more than two classes, with the softmax function and the cross-entropy loss.

# Check your understanding: part 3a

1. Why can the output of linear regression not be read as a probability?
2. A model gives $f_{\mathbf{w},b}(\mathbf{x}) = 0.2$ for a tumour. What is the probability that it is benign? What is $\hat{y}$ for $\tau = 0.5$?
3. With $\mathbf{w} = (2, -1)$ and $b = 1$, is the point $\mathbf{x} = (0, 3)$ classified as 1 or 0?
4. Why do we not use the squared error as the cost for logistic regression?
5. The gradient-descent updates look identical to those of linear regression. What is actually different?

# Part 3b: multiclass classification

# Where we are

In part 3a we classified into **two** classes:

- a linear score $z = \mathbf{w} \cdot \mathbf{x} + b$, turned into a probability by the sigmoid, $P(y = 1 \mid \mathbf{x}) = g(z)$;
- the logistic loss, convex and derived from maximum likelihood;
- gradient descent with the familiar gradient "error times input".

Part 3b handles $K \geq 3$ classes with the same ingredients:

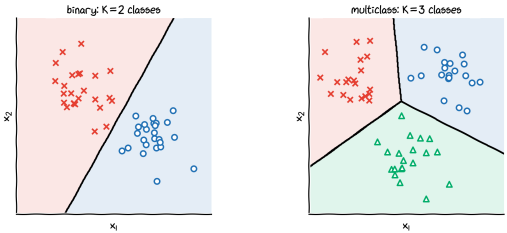
- **one score per class**, $z_k = \mathbf{w}_k \cdot \mathbf{x} + b_k$;
- the **softmax** turns the $K$ scores into $K$ probabilities that sum to 1 (the sigmoid's job for $K = 2$);
- the **cross-entropy** loss generalises the logistic loss, and the gradient is again error times input;
- for $K = 2$, everything reduces exactly to part 3a.

# Multiclass classification

# From two classes to many

Many questions have more than two possible answers:

- Which **digit** (0 to 9) is in this image?
- Which **species** is this flower?
- Which **blood type** does a patient have, given test results?
- Which **degree programme** will a student choose, given grades and interests?




- **Multiclass classification**: assign each example to exactly one of $K \geq 3$ classes.
- Not to be confused with **multi-label** classification, where an example can belong to several classes at once (e.g. the tags of a photo).

# Notation

Data set with $m$ examples, $n$ features and $K$ classes:

$$\mathcal{D} = \{(\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(m)}, y^{(m)})\}, \qquad \mathbf{x}^{(i)} \in \mathbb{R}^n, \qquad y^{(i)} \in \{1, 2, \dots, K\}.$$

In binary classification (part 3a) the model gave two probabilities:

$$\underbrace{P(y = 0 \mid \mathbf{x})}_{1 - f_{\mathbf{w},b}(\mathbf{x})} + \underbrace{P(y = 1 \mid \mathbf{x})}_{f_{\mathbf{w},b}(\mathbf{x})} = 1.$$

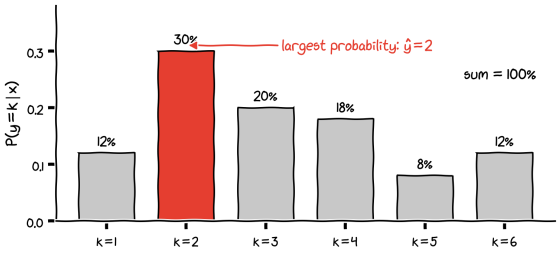
Now the model must give $K$ probabilities, one per class, that form a **probability distribution**:

$$P(y = k \mid \mathbf{x}) \geq 0 \quad \text{for } k = 1, \dots, K, \qquad \sum_{k=1}^{K} P(y = k \mid \mathbf{x}) = 1.$$

The method is called **softmax regression** or **multinomial logistic regression** (also: multinomial logit, maximum-entropy classifier).

# The softmax model

# Making a prediction




1. Compute all $K$ probabilities $P(y = k \mid \mathbf{x})$, $k = 1, \ldots, K$.
2. Predict the class with the **largest** probability:

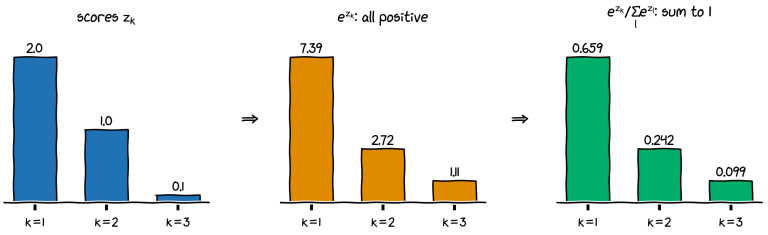
$$\hat{y} = \arg\max_k P(y = k \mid \mathbf{x}).$$

In the example $K = 6$, and $\hat{y} = 2$ because $P(y = 2 \mid \mathbf{x}) = 30\%$ is the largest.

# The softmax function

Turn $K$ arbitrary real numbers $z_1, \dots, z_K$ ("scores") into a probability distribution:

$$\text{softmax}(\mathbf{z})_k = \frac{e^{z_k}}{\sum_{l=1}^{K} e^{z_l}}, \qquad k = 1, \dots, K.$$




- Every output is positive, and $\sum_{k=1}^{K} \text{softmax}(\mathbf{z})_k = \left(e^{z_1} + \cdots + e^{z_K}\right) \Big/ \sum_l e^{z_l} = 1$.
- The largest score gets the largest probability. Softmax is a "soft" version of picking the maximum.

# The softmax regression model

Each class $k$ gets its own linear score, with its own weights and bias:

$$z_k = \mathbf{w}_k \cdot \mathbf{x} + b_k, \qquad P(y = k \mid \mathbf{x}) = \frac{e^{\mathbf{w}_k \cdot \mathbf{x} + b_k}}{\sum_{l=1}^{K} e^{\mathbf{w}_l \cdot \mathbf{x} + b_l}}, \qquad k = 1, \dots, K.$$

All parameters together: a weight matrix $\mathbf{W}$ with one **row per class**, and a bias vector $\mathbf{b}$:

$$\mathbf{W} = \begin{pmatrix} — \ \mathbf{w}_1 \ — \\ \vdots \\ — \ \mathbf{w}_K \ — \end{pmatrix} = \begin{pmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{K1} & w_{K2} & \cdots & w_{Kn} \end{pmatrix} \in \mathbb{R}^{K \times n}, \qquad \mathbf{b} = (b_1, \dots, b_K).$$

$w_{kj}$ is the weight of feature $j$ for class $k$. In total: $Kn + K = K(n + 1)$ parameters.

# Two classes: back to logistic regression

With $K = 2$ classes, called 0 and 1, divide numerator and denominator by $e^{z_1}$:

$$P(y = 1 \mid \mathbf{x}) = \frac{e^{z_1}}{e^{z_0} + e^{z_1}} = \frac{1}{1 + e^{z_0 - z_1}} = \frac{1}{1 + e^{-[(\mathbf{w}_1 - \mathbf{w}_0)\cdot\mathbf{x} + (b_1 - b_0)]}}.$$

With $\mathbf{w} = \mathbf{w}_1 - \mathbf{w}_0$ and $b = b_1 - b_0$ this is exactly the sigmoid model of part 3a:

$$P(y = 1 \mid \mathbf{x}) = g(\mathbf{w} \cdot \mathbf{x} + b) = f_{\mathbf{w},b}(\mathbf{x}), \qquad P(y = 0 \mid \mathbf{x}) = 1 - f_{\mathbf{w},b}(\mathbf{x}).$$

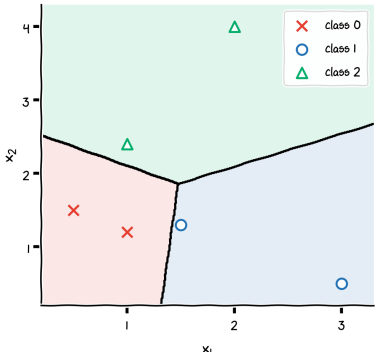
- Softmax regression **generalises** logistic regression.
- Only **differences** between the class scores matter: adding the same vector to every $\mathbf{w}_k$ (and the same number to every $b_k$) does not change any probability. Regularisation picks one of these equivalent solutions.

# Decision boundaries

Between classes $j$ and $k$ the boundary is where both are equally likely:

$$P(y = j \mid \mathbf{x}) = P(y = k \mid \mathbf{x}) \iff e^{z_j} = e^{z_k} \iff (\mathbf{w}_k - \mathbf{w}_j) \cdot \mathbf{x} + (b_k - b_j) = 0,$$

a straight line (a hyperplane in $n$ dimensions).




- Each pairwise line only matters where $j$ and $k$ are the two most likely classes.
- So the plane is divided into $K$ **convex regions**, one per class, with straight borders.
- Example: six points in three classes, fitted with scikit-learn (code later in the lecture).

# Training: the cross-entropy loss

# One-hot encoding

Write the label $y^{(i)}$ as a vector of length $K$ with a single 1:

$$y = 2,\ K = 3 \quad \longrightarrow \quad \mathbf{y} = (0,\ 1,\ 0), \qquad \text{entry } k: \quad \delta_{k,y} = \begin{cases} 1 & \text{if } k = y \\ 0 & \text{if } k \neq y \end{cases}$$

The model's output for the same example is also a vector of length $K$:

$$\mathbf{p} = (P(y = 1 \mid \mathbf{x}), \dots, P(y = K \mid \mathbf{x})), \qquad p_k = \text{softmax}(\mathbf{z})_k.$$

Training should make $\mathbf{p}$ close to the one-hot vector $\mathbf{y}$: a large probability for the true class.

# The cross-entropy loss

**Cost function of softmax regression**

$$J(\mathbf{W}, \mathbf{b}) = -\frac{1}{m} \sum_{i=1}^{m} \sum_{k=1}^{K} \delta_{k,y^{(i)}} \log p_k^{(i)} = -\frac{1}{m} \sum_{i=1}^{m} \log p_{y^{(i)}}^{(i)}$$

- For each example, only one term of the inner sum survives: **minus the log-probability of the true class**. This is the same idea as the logistic loss of part 3a, and again it follows from maximum likelihood.

Example with $\mathbf{p} = (0.659,\ 0.242,\ 0.099)$:

| true class | loss $-\log p_y$ |
|---|---|
| $y = 1$ | 0.42 (model fairly confident and right) |
| $y = 2$ | 1.42 |
| $y = 3$ | 2.31 (the model gave the true class only 10%) |

# Two classes: the binary cross-entropy

For $K = 2$ with classes 0 and 1:

$$\delta_{1,y} = y, \qquad \delta_{0,y} = 1 - y, \qquad p_1 = f_{\mathbf{w},b}(\mathbf{x}), \qquad p_0 = 1 - f_{\mathbf{w},b}(\mathbf{x}).$$

So the cross-entropy becomes

$$J = -\frac{1}{m} \sum_{i=1}^{m} \Big[ (1 - y^{(i)}) \log \left(1 - f_{\mathbf{w},b}(\mathbf{x}^{(i)})\right) + y^{(i)} \log f_{\mathbf{w},b}(\mathbf{x}^{(i)}) \Big],$$

which is exactly the logistic cost of part 3a. Softmax regression with its cross-entropy loss contains binary logistic regression as the special case $K = 2$.

# The gradient

For one example, with scores $z_k = \mathbf{w}_k \cdot \mathbf{x} + b_k$ and loss $L = -\log p_y$:

$$\frac{\partial L}{\partial z_k} = p_k - \delta_{k,y} \qquad \text{(predicted probability minus one-hot label: the error).}$$

Since $\partial z_k / \partial w_{kj} = x_j$ and $\partial z_k / \partial b_k = 1$, averaging over all examples gives

$$\frac{\partial J}{\partial w_{kj}} = \frac{1}{m} \sum_{i=1}^{m} \left(p_k^{(i)} - \delta_{k,y^{(i)}}\right) x_j^{(i)}, \qquad \frac{\partial J}{\partial b_k} = \frac{1}{m} \sum_{i=1}^{m} \left(p_k^{(i)} - \delta_{k,y^{(i)}}\right).$$

Once again: **error times input**, exactly as in Lecture 2 and part 3a.

Derivation of the first line: the softmax derivative is $\partial p_y / \partial z_k = p_y(\delta_{k,y} - p_k)$, so $\partial L / \partial z_k = -\frac{1}{p_y}\, p_y(\delta_{k,y} - p_k) = p_k - \delta_{k,y}$.

# Vectorised implementation

```
import numpy as np

def softmax(Z):
    Z = Z - Z.max(axis=1, keepdims=True)  # stability
    E = np.exp(Z)
    return E / E.sum(axis=1, keepdims=True)

def gradient_descent(X, y, K, lam=1.0,
                     alpha=0.5, n_iters=20000):
    m, n = X.shape
    W, b = np.zeros((K, n)), np.zeros(K)
    Y = np.eye(K)[y]                # one-hot, (m, K)
    for _ in range(n_iters):
        P = softmax(X @ W.T + b)  # (m, K)
        E = P - Y                 # errors
        W = W - alpha * (E.T @ X / m + lam / m * W)
        b = b - alpha * E.mean(axis=0)
    return W, b
```

- `X @ W.T + b` computes all $m \times K$ scores at once.
- `E.T @ X / m` is the gradient $\frac{1}{m}(\mathbf{P}-\mathbf{Y})^\top \mathbf{X}$ for all $w_{kj}$.
- With $\lambda = 1$ (Lecture 4) this reproduces scikit-learn's `LogisticRegression(C=1)` on the iris data to three decimals.

**Why subtract the maximum?** $e^{1000}$ overflows: the naive softmax of $(1000, 999, 998)$ gives `nan`. Since $\text{softmax}(\mathbf{z} - c) = \text{softmax}(\mathbf{z})$, subtracting $\max_k z_k$ changes nothing mathematically and gives $(0.665, 0.245, 0.090)$.

# Softmax regression in scikit-learn

# A small example with three classes

```
import numpy as np
from sklearn.linear_model import LogisticRegression

X_train = np.array([[0.5, 1.5], [1, 1.2], [1.5, 1.3],
                    [3, 0.5], [2, 4], [1, 2.4]])
y_train = np.array([0, 0, 1, 1, 2, 2])

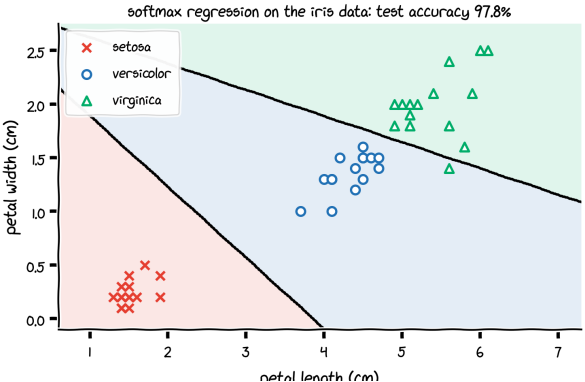
model = LogisticRegression()    # softmax for K > 2
model.fit(X_train, y_train)

model.predict(X_train)          # [0 0 1 1 2 2]
model.predict_proba(X_train[:2])
# [[0.609 0.185 0.206]
#  [0.522 0.309 0.168]]   each row sums to 1
model.coef_         # W, shape (3, 2): one row per class
# [[-0.639 -0.376] [ 0.627 -0.497] [ 0.012  0.873]]
model.intercept_    # b, shape (3,): [ 1.641 -0.002 -1.640]
```

These are the six points of the decision-boundary slide.

- For more than two classes, `LogisticRegression` uses softmax automatically.
- Older code passes `multi_class="multinomial"`. That option was removed in scikit-learn 1.8 and now raises an error: simply leave it out.
- `predict_proba` (not `predict_prob`) returns one row of $K$ probabilities per example.

# Real data: the iris flowers




150 iris flowers of three species (setosa, versicolor, virginica), collected by Edgar Anderson and made famous by R. A. Fisher (1936).

- Two features: petal length and petal width.
- 70% of the flowers for training, 30% for testing.
- Softmax regression gives three convex regions with straight borders, and **97.8%** test accuracy.

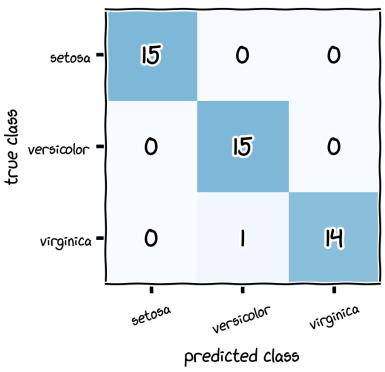
# The iris example in code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix

iris = load_iris()
X, y = iris.data[:, 2:], iris.target          # petal length and width; 3 species
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=0, stratify=y)

model = make_pipeline(StandardScaler(), LogisticRegression())
model.fit(X_train, y_train)
model.score(X_test, y_test)                              # 0.978: accuracy
confusion_matrix(y_test, model.predict(X_test))          # see the next slide
```

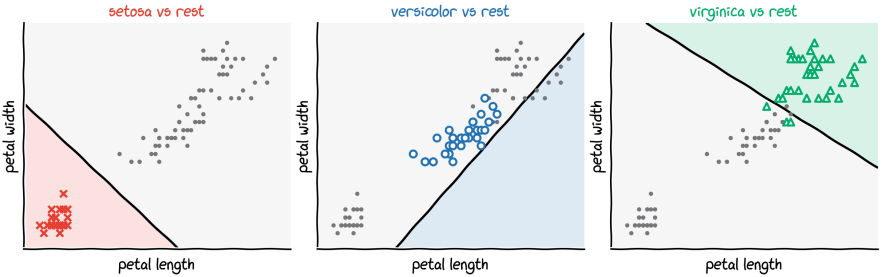
# Evaluating: the confusion matrix




Entry $(i, j)$ counts the test examples of **true** class $i$ that were **predicted** as class $j$.

- The diagonal holds the correct predictions: $15 + 15 + 14 = 44$ of 45, i.e. accuracy 97.8%.
- Off-diagonal entries show **which** classes are confused: here one virginica was predicted as versicolor. Setosa is never confused.
- With many classes, the confusion matrix is much more informative than the accuracy alone.

# An alternative: one-vs-rest




- Train $K$ separate **binary** classifiers: "class $k$" against "all other classes".
- Predict the class whose classifier is the most confident.
- Works with any binary classifier (`sklearn.multiclass.OneVsRestClassifier`). Softmax regression instead trains one model whose probabilities are consistent and sum to 1.

# Summary of part 3b

- **Multiclass classification**: $y \in \{1, \dots, K\}$; the model outputs a probability distribution over the $K$ classes, and predicts the most probable class.
- **Softmax**: $\text{softmax}(\mathbf{z})_k = e^{z_k} / \sum_l e^{z_l}$; **softmax regression** uses $z_k = \mathbf{w}_k \cdot \mathbf{x} + b_k$, with $K(n+1)$ parameters.
- For $K = 2$, softmax regression **is** logistic regression.
- **Decision regions** are convex, with straight borders $(\mathbf{w}_k - \mathbf{w}_j) \cdot \mathbf{x} + (b_k - b_j) = 0$.
- **Cross-entropy loss**: $J = -\frac{1}{m} \sum_i \log p^{(i)}_{y^{(i)}}$; gradient $\frac{1}{m} \sum_i (p^{(i)}_k - \delta_{k,y^{(i)}})\, x^{(i)}_j$: error times input.
- **Evaluation**: accuracy and the **confusion matrix**.

# Check your understanding: part 3b

1. Compute $\text{softmax}(1, 1, 1)$. Which class is predicted?
2. How many parameters does softmax regression have for $n = 4$ features and $K = 3$ classes?
3. The model gives $\mathbf{p} = (0.7, 0.2, 0.1)$ and the true class is $y = 2$. What is the loss for this example?
4. What does an off-diagonal entry of 5 in row "cat", column "dog" of a confusion matrix mean?

# Wrapping up

# Lecture 3 at a glance

| | two classes (3a) | $K$ classes (3b) |
|---|---|---|
| labels | $y \in \{0, 1\}$ | $y \in \{1, \dots, K\}$, one-hot vector $\mathbf{y}$ |
| scores | $z = \mathbf{w} \cdot \mathbf{x} + b$ | $z_k = \mathbf{w}_k \cdot \mathbf{x} + b_k, \quad k = 1, \dots, K$ |
| probabilities | $P(y = 1 \mid \mathbf{x}) = g(z) = \frac{1}{1+e^{-z}}$ | $P(y = k \mid \mathbf{x}) = \frac{e^{z_k}}{\sum_l e^{z_l}}$ |
| prediction | $\hat{y} = 1$ if $g(z) \geq \tau$ | $\hat{y} = \arg\max_k P(y = k \mid \mathbf{x})$ |
| loss | $-y \log g(z) - (1 - y) \log(1 - g(z))$ | $-\log P(y = y^{(i)} \mid \mathbf{x})$ |
| gradient | error times input | error times input, for every class |
| scikit-learn | `LogisticRegression` | `LogisticRegression` (softmax automatically) |

**Next lecture**: overfitting, regularisation and model selection, for regression and classification alike.